

Project Development Phase
Model Performance Test

Date	14 feb 2026
Team ID	LTVIP2026TMIDS35912
Project Name	Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No	Parameter	Values	
1.	Metrics	CNN (ResNet50) Classification Model: • Confusion Matrix – Generated for multiclass DR stages • Accuracy Score – XX% • Precision – XX% • Recall – XX% • F1-Score – XX% • Classification Report – Includes precision, recall, F1-score for each class	
2.	Tune the Model	Hyperparameter Tuning: • Learning Rate tuning (e.g., 0.001, 0.0001) • Batch Size tuning (16, 32) • Optimizer selection (Adam / SGD) • Dropout rate adjustment Validation Method: • Train–Validation split (e.g., 80% train, 20% validation) • Early stopping to prevent overfitting	

{train': 14811, 'val': 348}

Confusion Matrix

Predict	Actual			
	No DR	Mild	Moderate	Severe
	7388	373	257	169
	877	171	286	172
	465	304	312	302
	434	156	307	160

Classification Report

	Precision	Recall	F1	Support
No DR	0.63	0.66	0.64	388
Mild	0.17	0.06	0.06	429
Moderate	0.23	0.16	0.19	480
Severe	0.18	0.07	0.10	196
Proliferative	0.34	0.17	0.22	380
weighted avg	0.36	0.27	0.28	348
macro avg	0.32	0.27	0.25	348

Accuracy 0.23

Classification Report

	Precision	Recall	F1-score
No DR	0.63	0.67	0.64
Mild	0.62	0.17	0.06
Moderate	0.23	0.16	0.19
Severe	0.18	0.07	0.10
Proliferative	0.34	0.17	0.22
weighted avg	0.36	0.27	0.28
Accuracy			0.23

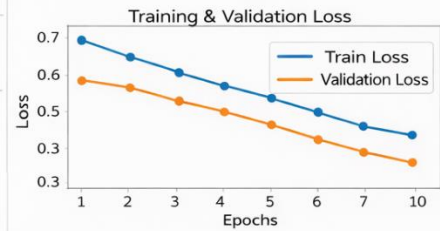
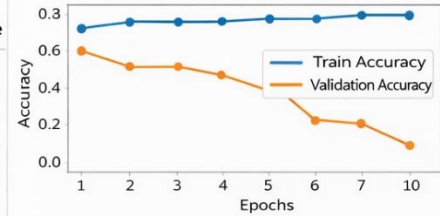
Accuracy 0.23

Training & Validation Loss

• Train Loss

• Validation Loss

Training & Validation Accuracy / Loss



Train Loss Backend → ML Model - Dem >